JUMIA NIGERIA RECONNAISSANCE

UCU Data Mining and Wrangling | Group 2 | Verified 13 October 2026

Purpose: confirm access rules, page structure and a responsible collection plan before scraping.

VERIFIED CATEGORY PAGES

Phones & Tablets https://www.jumia.com.ng/phones-tablets/

Computing https://www.jumia.com.ng/computing/

Electronics https://www.jumia.com.ng/electronics/

ROBOTS.TXT AND ETHICS

robots.txt returned HTTP 200. Its public-site guidance permits clearly identified bots with an owner/contact when they stay below 200 requests per minute.

The User-Agent identifies DSC2107-UCU-Group-2 and states that contact is supplied at runtime; this document does not print the email address.

Private/customer/cart paths, site search, tracking parameters, filters and crawl traps remain disallowed. Standard category pages and ?page=N pagination were checked as allowed.

TECHNICAL APPROACH

Product data is server-rendered HTML. Each successful category page exposed 40 main catalogue cards, so Requests plus BeautifulSoup is sufficient. Playwright and private/mobile endpoints are unnecessary.

Main card: article.prd._fb.col.c-prd. Stable ID and brand: a.core[data-gtm-id] and data-gtm-brand. Name, prices, discount, rating and reviews use h3.name, div.prc, div.old, div.bdg._dsct, div.stars and div.rev.

PAGINATION AND TIMESTAMPS

Next-page links use ?page=2#catalog-listing; the HTTP request needs only ?page=2 and then increasing page numbers.

Every observation receives a genuine timezone-aware Africa/Nairobi timestamp and collection date. Missing values remain missing rather than becoming zero.

REQUIRED OBSERVATIONS

Stable product ID, product name, brand, raw current and old prices, displayed discount, rating, review count, assigned category, canonical URL, NGN currency, source site, scraped_at and scrape_date.

Cross-category repeats are deduplicated by stable ID while category membership is preserved.

RATE AND REQUEST ESTIMATE

Requests are sequential with a fixed three-second delay - about 20 requests per minute, well below the published 200-per-minute ceiling. The run also keeps a request counter and a 1,000-request safety limit.

Sample estimate: robots.txt plus one page from each category = about 4 requests and about 120 cards. Daily estimate: robots.txt plus about five pages per category = about 16 requests for roughly 600 cards, with limited retries if needed.

KNOWN CHALLENGES

Old price, discount, rating and review count can be genuinely absent. Products can overlap categories, promotions can change HTML, and temporary network failures can occur.

A Computing request returned a preserved HTTP 503 maintenance page; a later polite request returned HTTP 200. Production logic therefore uses bounded retries and controlled back-off, but stops on CAPTCHA, HTTP 403, HTTP 429 or explicit denial.

EVIDENCE AND DECISION

Evidence is stored under reconnaissance/evidence/: robots.txt, HTTP metadata, timestamps, response sizes and SHA-256 hashes. Large HTML captures remain local and outside Git.

Decision: use a polite, sequential Requests/BeautifulSoup scraper and validate a 100-product sample before any full daily run.